

Q1. What must be checked before these tables can be used?

Before joining or analysing the two workbooks, I first checked their structure because I did not want to assume that every row represented a sample. The OLD workbook contained 3,441 rows and 18 columns, while the raw NEW workbook contained 3,442 rows and 70 columns. One row in the NEW workbook had no Unique ID, but inspection showed that this was not a sample with a missing identifier. It was a metadata row containing measurement units. I preserved this row separately instead of treating it as sample data. After doing this, the NEW sample table also contained 3,441 records.

Unique ID was used as the candidate key. Both sample tables had zero missing and zero duplicate identifiers, which was important before attempting a one-to-one join.

I selected Cu, Zn, U, Fe and Mn for the chemical audit. One issue that stood out was censored data: OLD U contained 11 censored entries, while NEW Mn contained 34 censored entries. These values should not simply be converted to zero or deleted because the censoring itself contains information about the measurement.

There was also a metadata limitation. The NEW workbook documents Cu, Zn, U and Mn in ppm and Fe in %, but equivalent units are not provided in the OLD workbook. Because of this, I kept the original values and recorded the uncertainty in `data_dictionary.csv` and `audit.csv` rather than assuming that OLD and NEW measurements are directly comparable.

Q2. Can every joined record be traced to its sources?

Before joining, I checked the uniqueness of Unique ID in both datasets. Both contained 3,441 unique sample identifiers and no duplicate keys. I then performed an outer one-to-one join so that unmatched records would remain visible rather than being silently discarded.

The join produced 3,441 matched records, with zero OLD-only and zero NEW-only records. This reconciles with the original counts: all 3,441 OLD identifiers have a corresponding NEW identifier and vice versa. Although this is a clean result, I still kept the merge status in the joined output because traceability should not depend on assuming that every join will always be complete.

For the five-sample trace, I used the student-number-derived seed of 12016 rather than manually selecting convenient records. The trace file records the source workbook, source row, join status, Cu and Zn values, and coordinate fields for each selected identifier.

The coordinates need additional care. OLD coordinates are labelled NAD83, whereas NEW coordinates are labelled NAD27. I therefore did not treat numerical differences between the two coordinate pairs as evidence that a sample moved. A documented coordinate transformation and fuller CRS information would be needed before making that comparison.

The same caution applies to chemical results. The NEW units are documented, but OLD units and analytical-method information are incomplete. The joined database can therefore

establish record relationships, but the join alone does not prove analytical equivalence between measurements.

Q3. Which unusual records deserve investigation, and why?

The supplied Cu–Zn–U baseline was run using complete numeric cases. It produced 3,440 eligible records, with one excluded from the feature matrix. StandardScaler was used before Isolation Forest, with a contamination setting of 0.05 and seed 12016.

For my alternative analysis, I added Mn to the Cu–Zn–U feature set. My reason was to test whether an additional geochemical variable would reveal unusual multivariate behaviour that was not visible from Cu, Zn and U alone. This choice also had a cost: the eligible count decreased from 3,440 to 3,406 because Mn introduced additional incomplete/non-numeric cases.

The comparison produced genuine classification disagreements. Five particularly clear examples were 013M781525, 014D783063, 014D781076, 014D783153 and 013M783296. All five were classified as normal by the baseline but anomalous after Mn was introduced.

The size of some rank changes was substantial. Sample 013M781525, for example, moved from baseline rank 3422 to alternative rank 78 and had a NEW Mn concentration of 7,308 ppm. 013M783296 moved from rank 2601 to rank 39, with Mn of 8,108 ppm.

I marked these records as **query**, not exclude. Their behaviour gives a good reason to investigate the assay and source information, but an Isolation Forest result is not proof of an incorrect measurement. Excluding them simply because a model considers them unusual could remove genuine geological information. The lack of documented OLD units and analytical-method comparability also prevents me from using OLD–NEW differences as definitive evidence of error.

Q4. Is the workflow reproducible, and which choice do you defend?

I kept the supplied Cu–Zn–U analysis as my baseline and used Cu–Zn–U–Mn as an alternative rather than replacing the baseline completely. This made the effect of my analytical choice visible. One clear numerical consequence was the reduction in eligible records from 3,440 to 3,406 after Mn was added. More importantly, sample 013M783296 changed from rank 2601 to rank 39. This showed me that anomaly rankings can depend strongly on which variables are included.

For that reason, I would use the original baseline as the reproducible reference and the Mn analysis as supporting evidence for deciding which records deserve investigation. I would not treat either model as an automatic rule for deleting samples.

Reproducibility was addressed by using student number 2539990 and seed 12016, recording the software versions and model configuration, and generating SHA-256 hashes for the input

workbooks in run_manifest.json. Required tables were exported rather than manually copied from notebook output.

One unresolved limitation remains the incomplete analytical metadata for the OLD workbook, particularly units, methods and detection limits. This prevents a fully supported direct comparison of OLD and NEW assay values.